MIL-C-24643/30C
14 March 1994
SUPERSEDED
MIL-C-24643/30B
3 October 1986

MILITARY SPECIFICATION SHEET

CABLE, ELECTRICAL, TYPE LS1SWU (INCLUDING VARIATION LS1SWA)

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the product described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-C-24643.

This cable can be manufactured in two variations: LS1SWU (unarmored), and LS1SWA (armored).

REQUIREMENTS:

Qualification required.

Construction (watertight)

First - Copper conductor, uncoated, ASTM B 286, size 22-7. maximum diameter 0.033 inch.

Second - Separator may be used at manufacturer's option where required to provide free-stripping insulation.

Third - Crosslinked polyethylene insulation. Nominal wall thickness 0.048 inch; wall thickness may vary from nominal as necessary, in order for completed cable to meet the specified electrical requirements.

Fourth - Braided shield of AWG no. 36 uncoated copper. Braid angle of 30 to 35 degrees; minimum coverage of 85 percent.

* Fifth - Shield insulation of two polyester tapes, type G in accordance with MIL-I-631, sealed, An alternate shield insulation may consist of one tape, type G of MIL-I-631, plus an extruded jacket of a suitable transparent material, minimum average thickness of the transparent material 0.003 inch. The standard identification code shall be applied by method 2 on the inner tape.

* Sixth - The required number of singles (see table I) cabled diameter of the layer. Cabling sequence shall be consecutive, starting with no. 1, from the center outward. Fillers shall be used as necessary to make a firm, well rounded assembly.

Seventh- Binder tape, applied helically with overlap.

Construction variant for type LS1SWU

* Eighth - Cross-linked polyolefin jacket (see table I for thickness).

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Construction variant for type LS1SWA8

Eighth - Cross-linked polyolefin jacket (see table I for thickness).
Ninth - Braided metal armor.

* TABLE I. Details.

Militar 1/ part no. M24643/3	Size 2/	Number of conductors	Cable jacket thickness min avg (inch)	Overall diameter (inches)			
				TYPE LS1SWU		TYPE LS1SWA	
				min	max	min	max
-01UN	2	2	0.038	0.430	0.455	0.480	0.505
-02UN	14	14	.050	.825	.870	.875	.925
-03UN	20	20	.050	.970	1.030	1.020	1.080
-04UN	30	30	.063	1.135	1.200	1.185	1.250

1/ Unarmored. When armor option is required, see MIL-C-24643 for configuration of part number.
2/ Type LS1SWU and LS1SWA.

EXAMINATION AND TESTS:	Requirements
Basic electrical:	
Conductor Resistance - ohms/1000 feet at 25°C, maximum	17.2
Voltage withstand - volts, root mean square, minimum	
Conductor to shield	3000
Shield to shield	500
Insulation resistance - Megohms/1000 feet minimum	
Conductor to shield	500
* shield to shield	100
Conductor and shield continuity	No failure
Group A:	
Visual and dimensional	No failure
Hydrostatic (open end) - leakage at 300 lb/in^2, 2 hours, in^3, maximum	1.0
Capacitance - 1 kilohertz (KHz), pF/ft, maximum	28
Characteristic impedance - at 1MHz, ohms	61 ± 3
Group B:	
* Cross-linked proof test (percent, maximum)	
Insulation	50
Jacket (When tested at 200°C)	50
Drip - 95 ± 1°C	Zero
Tear - pounds per inch thickness, minimum (ASTM D 470)	35
Physicals (unaged)	
Insulation	
Tensile strength - lb/in^2, minimum	1800
Elongation - percent, minimum	250
Jacket (cable)	
Tensile strength - lb/in^2, minimum	1300
Elongation - percent, minimum	160

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EXAMINATION AND TESTS (continued):	Requirements
Group C:	
Physicals (aged)	
Insulation	
Air oven	
Tensile strength - percent of unaged, minimum	80
Elongation - percent of unaged minimum	80
Jacket (cable)	
Air oven	
Tensile strength - percent of unaged, minimum	60
Elongation - percent of unaged minimum	60
Hot oil immersion	
* Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged minimum	50
Shrinkage	No failure
Heat distortion - percent of unaged, maximum	30
* Permanence of printing (jacket) - (types LS1SWU only) cycles, minimum	125
* Cable sealant removability	No failure
Armor (type LS1SWA only) - conformance to material, construction and coverage	No failure
Shield - conformance to material, construction and coverage	No failure
Group D:	
Flame propagation (cable)	No failure

QUALIFICATION INSPECTION:

* Qualification inspection shall include basic electrical, all of groups A, B, C and D, plus the following:

	Requirements
Aging and compatibility (cable) (125 ± 5°C)	No failure
Abrasion resistance (jacket) - scrapes, minimum	75
Acid gas equivalent - percent, maximum	
Jacket	2
Fillers	2
Insulation	18
* Shield insulation	2
Halogen content - percent, maximum	
Jacket	0.2
Fillers	0.2
Immersion (jacket)	
Tensile strength - percent of unaged, minimum	50
Elongation - percent of unaged, minimum	50
Smoke index, maximum	
Jacket	25
* Fillers	45
Insulation	45

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QUALIFICATION INSPECTION(continued):

	Requirements
Toxicity index, maximum	
Jacket	5
Fillers	5
Insulation	1.5
Durometer (jacket) - (type A) hardness, minimum	80
Weathering (jacket)	No failure

UNIT ORDERING LENGTHS: 1000 feet nominal

NOTE: The margins of this specification are marked with asterisks to indicate where changes (additions, modifications, corrections, deletions) from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous issue.

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